

AAA Authentication Using TACACS+, RADIUS, and Centralized Wireless Management

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Packet Tracer Version Used: v9.0.0.0810

Download: <https://www.netacad.com/articles/news/download-cisco-packet-tracer>

Objectives

- Implement TACACS+ for administrator authentication.
- Configure RADIUS for wireless administrator authentication.
- Deploy a Wireless LAN Controller (WLC) for centralized wireless management.
- Segment the network using VLANs.
- Provide Internet connectivity.

Project Overview

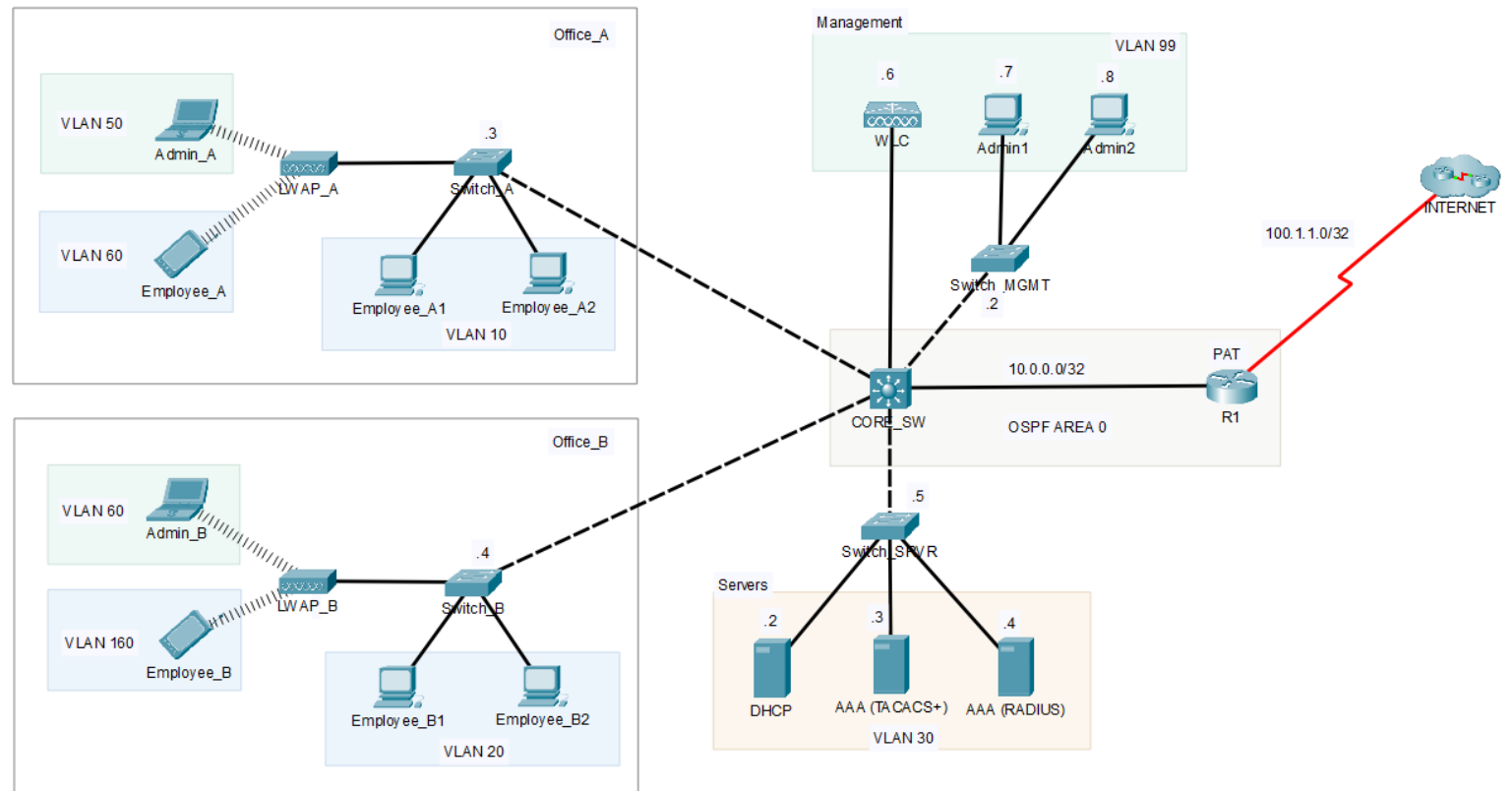
This project demonstrates a secure enterprise network integrating centralized wireless management and AAA authentication. The network consists of two office VLANs, a server network, and a dedicated management network.

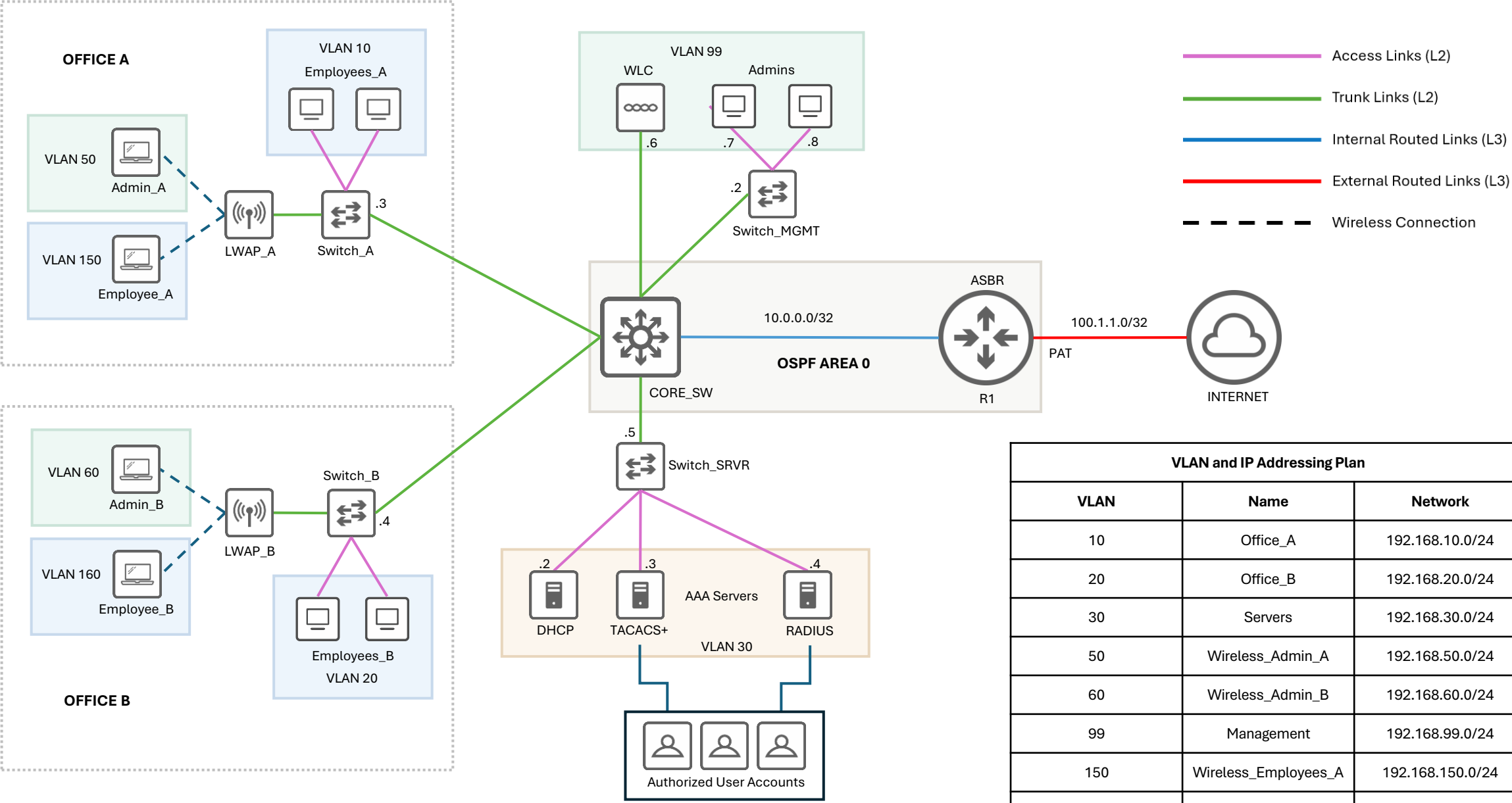
Wireless access points are centrally managed by a Wireless LAN Controller (WLC).

Network device administration is authenticated through a TACACS+ server, while administrative wireless users authenticate through a RADIUS server. Employee wireless users connect using a pre-shared key (PSK)

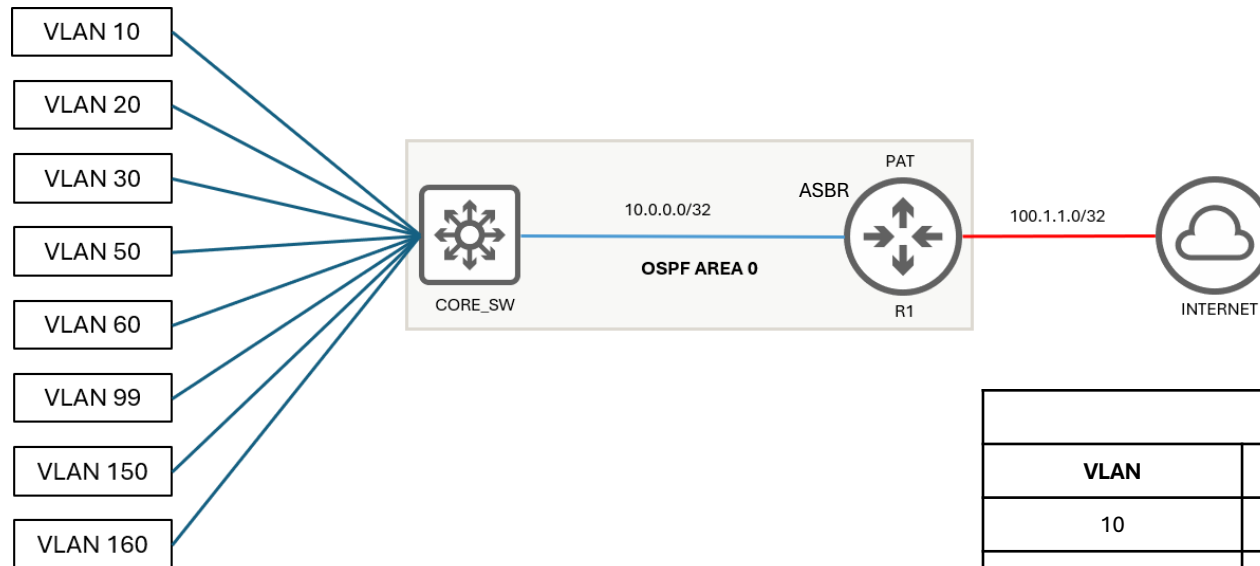
Network Topology

- Core Switch performs Inter-VLAN Routing
- WLC manages all LWAPs
- Centralized DHCP
- AAA Servers authenticate users
- Internet via PAT





Routing & Connectivity



- Inter-VLAN routing performed by CORE_SW
- OSPF Area 0 for dynamic routing
- PAT configured on R1
- Internet connectivity for internal users

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 100.1.1.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.255.255.2/32 is directly connected, Loopback0
    100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.1.1.0/30 is directly connected, Serial0/0/0
L       100.1.1.1/32 is directly connected, Serial0/0/0
O       192.168.10.0/24 [110/2] via 10.0.0.1, 00:00:11, GigabitEthernet0/0
O       192.168.30.0/24 [110/2] via 10.0.0.1, 00:00:11, GigabitEthernet0/0
O       192.168.50.0/24 [110/2] via 10.0.0.1, 00:00:11, GigabitEthernet0/0
O       192.168.60.0/24 [110/2] via 10.0.0.1, 00:00:11, GigabitEthernet0/0
O       192.168.99.0/24 [110/2] via 10.0.0.1, 00:00:11, GigabitEthernet0/0
O       192.168.150.0/24 [110/2] via 10.0.0.1, 00:00:11, GigabitEthernet0/0
O       192.168.160.0/24 [110/2] via 10.0.0.1, 00:00:11, GigabitEthernet0/0
S*      0.0.0.0/0 [1/0] via 100.1.1.2
```

CONFIGURED ON CORE_SW				
VLAN	Name	Network	Gateway (SVI)	Public IP Address
10	Office_A	192.168.10.0/24	192.168.10.1	100.1.1.1 (PAT)
20	Office_B	192.168.20.0/24	192.168.20.1	
30	Servers	192.168.30.0/24	192.168.30.1	
50	Wireless_Admin_A	192.168.50.0/24	192.168.50.1	
60	Wireless_Admin_B	192.168.60.0/24	192.168.60.1	
99	Management	192.168.99.0/24	192.168.99.1	
150	Wireless_Employees_A	192.168.150.0/24	192.168.150.1	
160	Wireless_Employees_B	192.168.160.0/24	192.168.160.1	

DHCP Services

- Centralized DHCP Server
- Server Separate scope per VLAN
- DHCP Relay using `ip helper-address` command for SVIs on CORE_SW

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Wireless_Admin_A	192.168.50.1	0.0.0.0	192.168.50.11	255.255.255.0	245	0.0.0.0	0.0.0.0
Wireless_Admin_B	192.168.60.1	0.0.0.0	192.168.60.11	255.255.255.0	245	0.0.0.0	0.0.0.0
Wireless_Employees_A	192.168.150.1	0.0.0.0	192.168.150.11	255.255.255.0	245	0.0.0.0	0.0.0.0
Wireless_Employees_B	192.168.160.1	0.0.0.0	192.168.160.11	255.255.255.0	245	0.0.0.0	0.0.0.0
MGMT_LWAP	192.168.99.1	0.0.0.0	192.168.99.11	255.255.255.0	245	0.0.0.0	192.168.99.6
Pool_B	192.168.20.1	0.0.0.0	192.168.20.11	255.255.255.0	245	0.0.0.0	0.0.0.0
Pool_A	192.168.10.1	0.0.0.0	192.168.10.11	255.255.255.0	245	0.0.0.0	0.0.0.0

SSH Configuration

SSH & AAA Configuration Summary					
Device	Management IP	Management VLAN	SSH	AAA	TACACS+ Server
CORE_SW	All L3 interface IPs	99	SSHv2	Enabled	192.168.30.3
Router (R1)	All L3 interface IPs	X			
Switch_MGMT	192.168.99.2	VLAN 99			
Switch_A	192.168.99.3				
Switch_B	192.168.99.4				
Switch_SRVR	192.168.99.5				

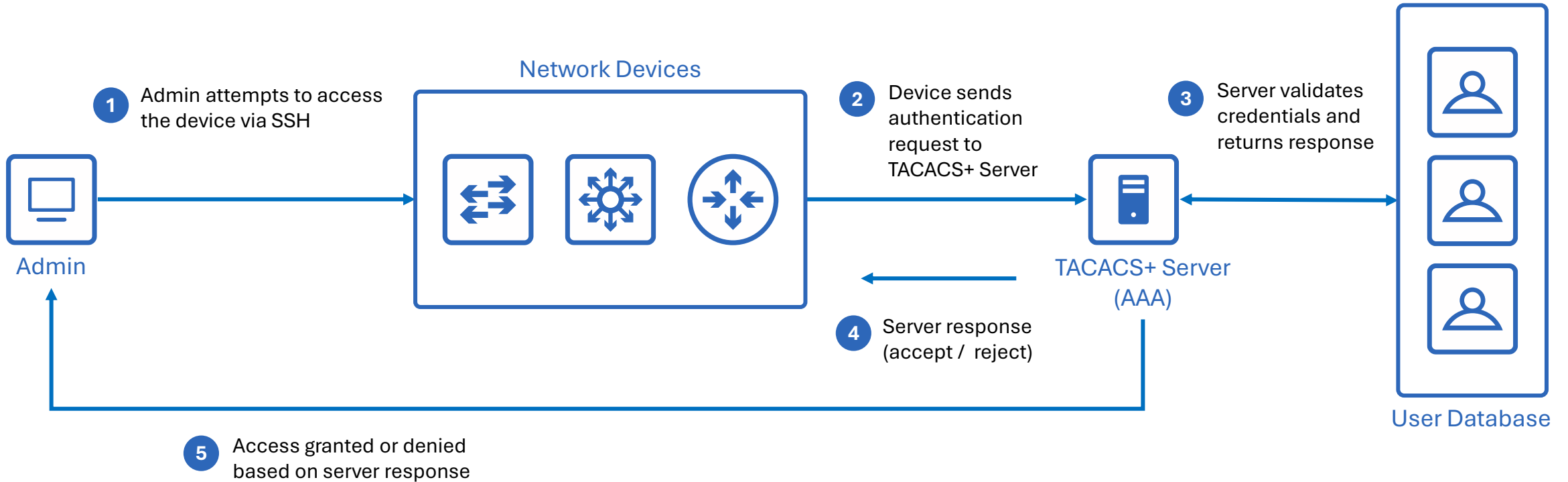
COMMON SSH CONFIGURATION							
SSH Version	RSA Key Length	Remote Access	Domain Name	Local Administrative Account	Enable Secret	AAA Authentication	VTY Lines
SSHv2	2048-bit	SSH only	company.local	superadmin	Configured	TACACS+ with local fallback	0–15

Management Security

SSH MANAGEMENT ACCESS (ALL NETWORK DEVICES)				
ACL Name	Permitted Network	User Group	Applied To	Purpose
SSH_MGMT	192.168.99.0/24	Management VLAN	VTY Lines (0–15)	Only trusted admin networks can SSH to network devices.
	192.168.50.0/24	Admin WLAN (Office A)		
	192.168.60.0/24	Admin WLAN (Office B)		

PROTECT MANAGEMENT VLAN (CORE SWITCH)				
ACL Name	Denied Network	Destination	Applied To	Purpose
BLOCK-MGMT	192.168.10.0/24	Management VLAN (99)	VLAN 10 SVI	Block Unauthorized users from having access to Management VLAN
	192.168.20.0/24		VLAN 20 SVI	
	192.168.150.0/24		VLAN 150 SVI	
	192.168.160.0/24		VLAN 160 SVI	

Authentication Flow (TACACS+)



TACACS+ AAA Server Configuration

The screenshot shows the AAA (TACACS+) configuration window. The 'Services' tab is selected, and the 'AAA' service is enabled. The 'Radius Port' is set to 1645. Under 'Network Configuration', a table lists six clients, all with 'Tacacs' as the 'Server Type' and 'aj123' as the 'Key'. A red box highlights this table. Below it, the 'User Setup' section shows a table with two users: 'Andres' and 'Tadeo', both with passwords ending in '123'. A green box highlights this table. Arrows point from these boxes to explanatory text on the right.

AAA (TACACS+)

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA**
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

AAA

Service ☒ On ☐ Off Radius Port 1645

Network Configuration

Client Name Client IP Secret ServerType Tacacs

	Client Name	Client IP	Server Type	Key
1	Router	10.0.0.2	Tacacs	aj123
2	CORE_SW	192.168.30.1	Tacacs	aj123
3	Switch_MGMT	192.168.99.2	Tacacs	aj123
4	Switch_A	192.168.99.3	Tacacs	aj123
5	Switch_B	192.168.99.4	Tacacs	aj123
6	Switch_SRVR	192.168.99.5	Tacacs	aj123

Add Save Remove

User Setup

Username Password

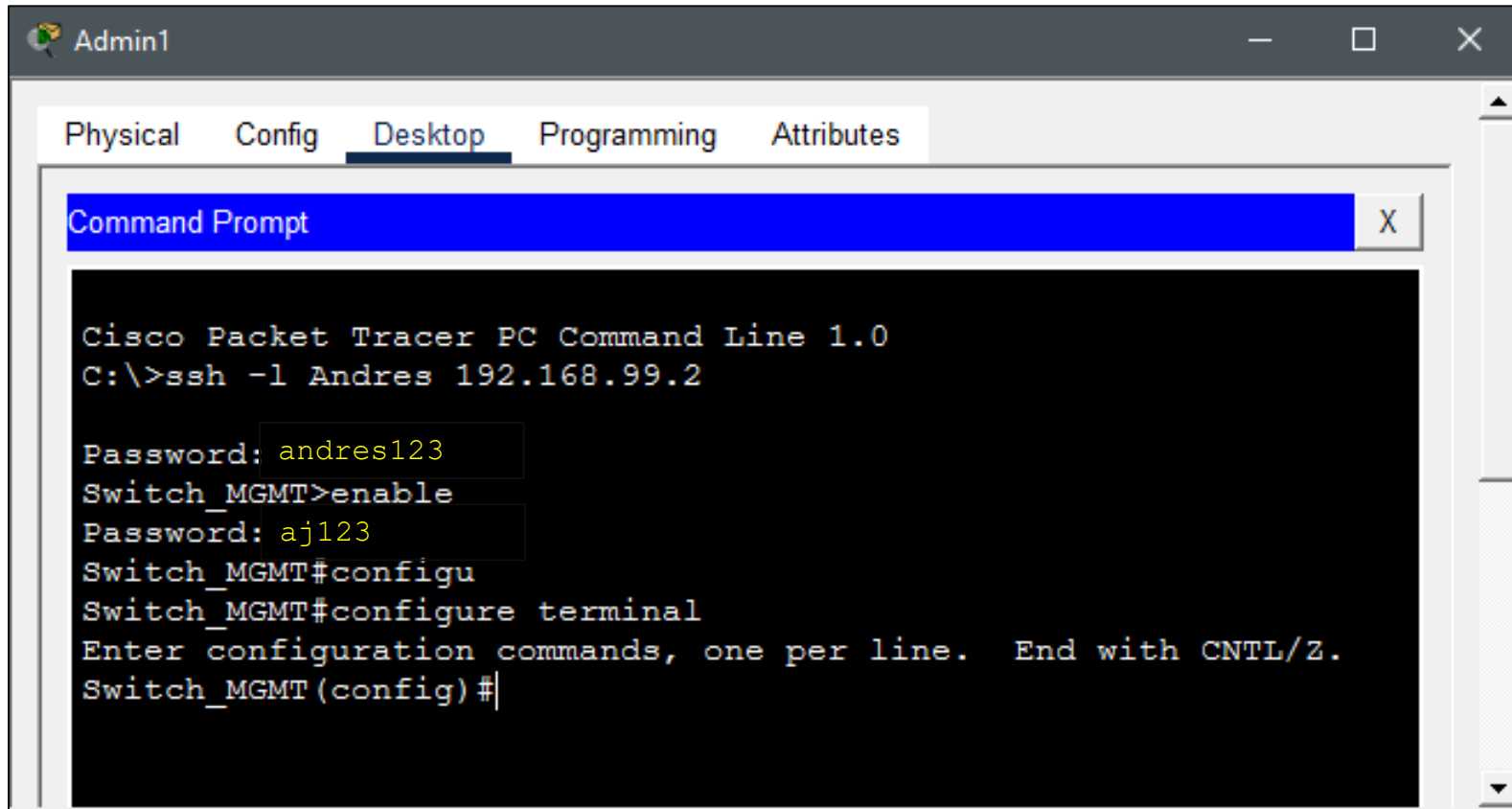
	Username	Password
1	Andres	andres123
2	Tadeo	tadeo123

Add

- AAA service enabled for TACACS+ authentication.
- Router and switches are registered as TACACS+ clients.
- Shared secret key "**aj123**" is configured for all clients.

User accounts are created for administrator authentication

SSH Verification with TACACS+



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'Admin1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The command prompt displays the following sequence of commands and responses:

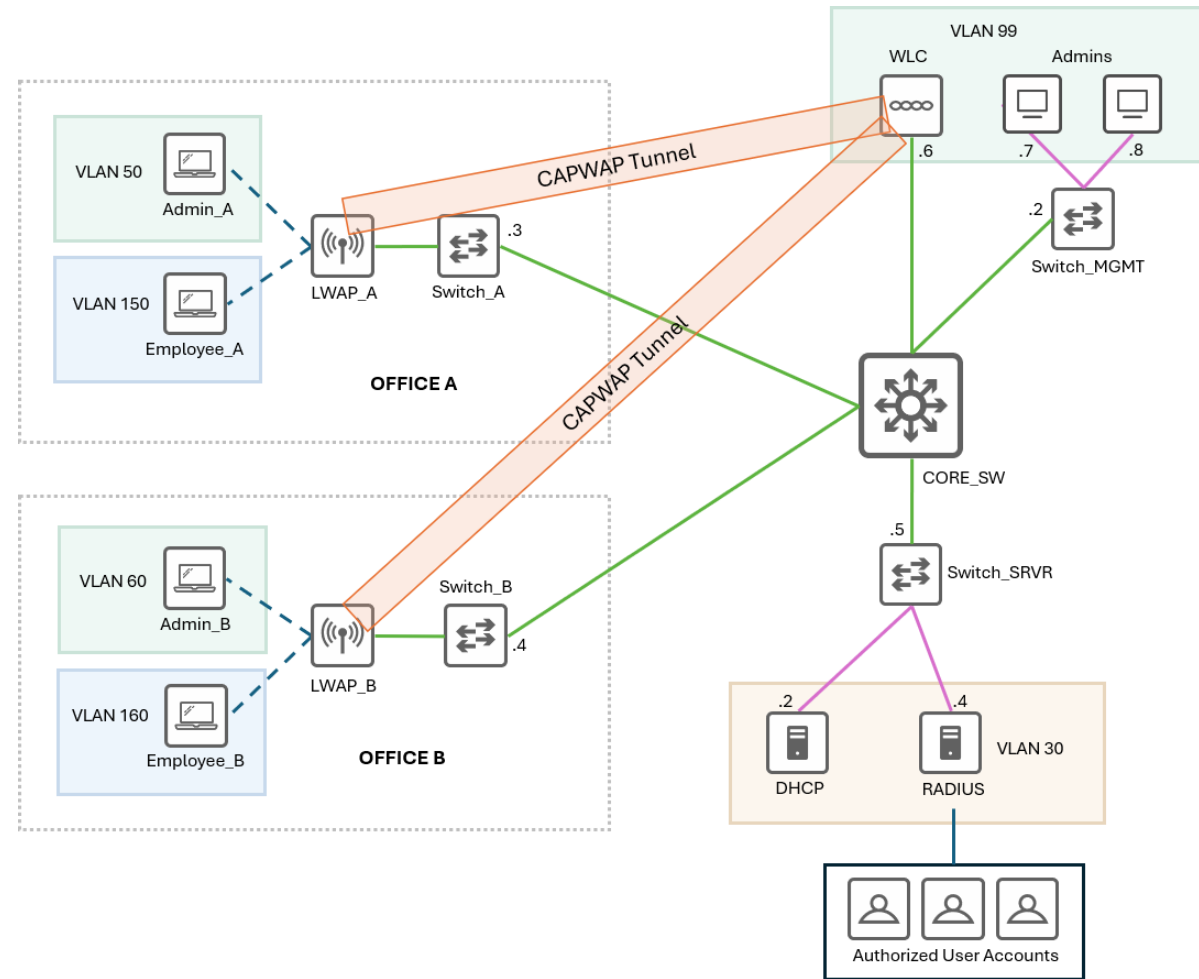
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l Andres 192.168.99.2

Password: andres123
Switch_MGMT>enable
Password: aj123
Switch_MGMT#configu
Switch_MGMT#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch_MGMT(config) #
```

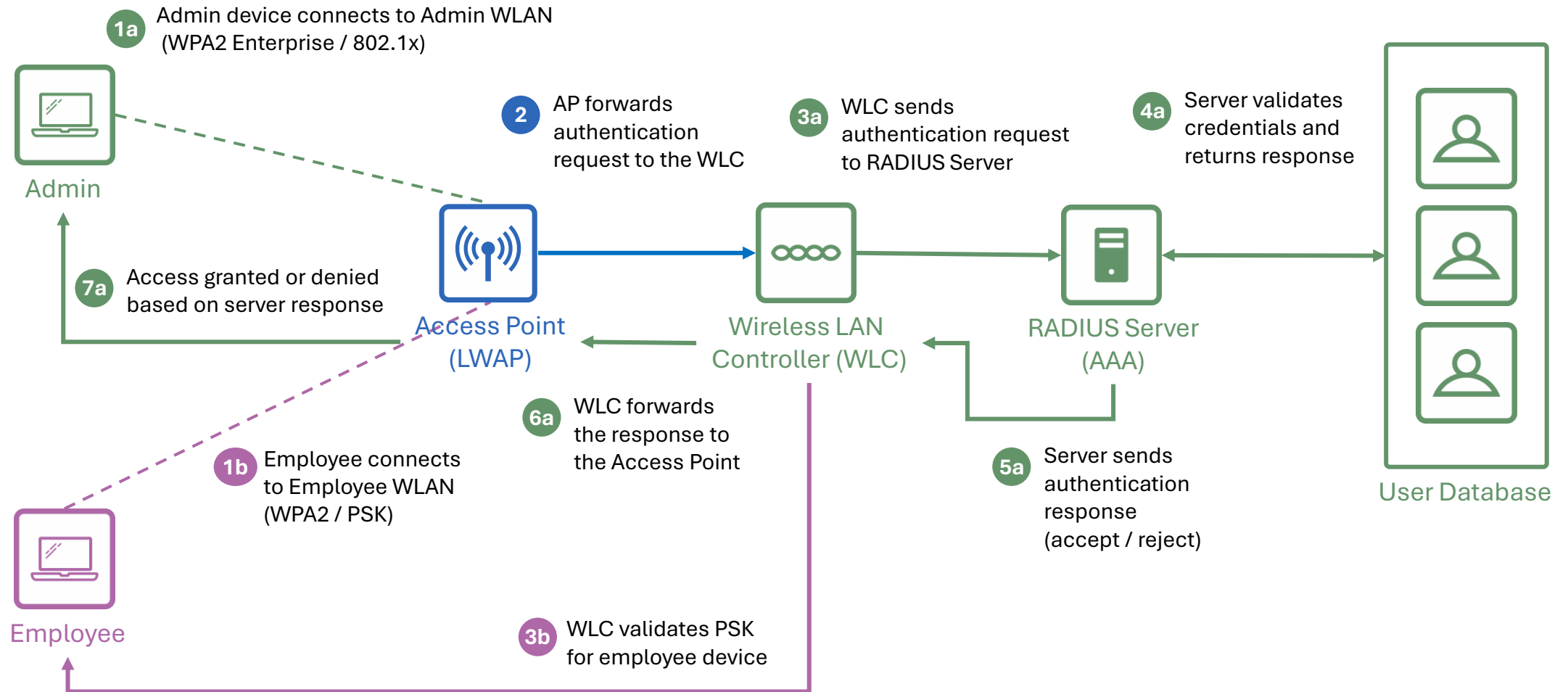
Wireless Infrastructure

Centralized Wireless Management

- Lightweight Access Points (LWAPs) centrally managed by the WLC via CAPWAP tunnels
- Separate wireless networks for administrators and employees
- WLC is configured by Network Admins
- User Authentication is handled by the RADIUS Server
- DHCP Server Provides IP Addressing Services for LWAPs and Wireless End Users



Authentication Flow (RADIUS)



RADIUS AAA Server Configuration

The screenshot shows the 'AAA (RADIUS)' configuration window with the 'Services' tab selected. The 'Service' is set to 'On' and the 'Radius Port' is '1645'. Under 'Network Configuration', a table lists a client named 'WLC' with IP '192.168.99.6', 'Radius' server type, and key 'aj123'. Below this, the 'User Setup' section contains a table with two users: 'Andres' (password 'andres123') and 'Tadeo' (password 'tadeo123').

Client Name	Client IP	Server Type	Key
1 WLC	192.168.99.6	Radius	aj123

	Username	Password
1	Andres	andres123
2	Tadeo	tadeo123

- AAA service enabled for RADIUS authentication.
- Wireless LAN Controller (WLC) is registered as the RADIUS client.
- Shared secret key "**aj123**" is configured between the WLC and RADIUS server.

User accounts are created for wireless user authentication.

Wireless LAN Controller Configuration

CONTROLLER INFORMATION	
Device Name	WLC_MGMT
Management VLAN	99
Management IP Address	192.168.99.6
Username	admin
Password	aj123.A
RADIUS Server	192.168.30.4 / Port 1645
DHCP Server	192.168.30.2
Web Login Page	https://192.168.99.6

CONTROLLER INTERFACES			
Interface	VLAN	IP Address	Purpose
Management	99	192.168.99.6	Controller management and AP control
Admin_A	50	192.168.50.10	Office A administrative wireless clients
Admin_B	60	192.168.60.10	Office B administrative wireless clients
Employees_A	150	192.168.150.10	Office A employee wireless clients
Employees_B	160	192.168.160.10	Office B employee wireless clients

Wireless LAN Controller Configuration

WLAN CONFIGURATION					
SSID	VLAN	Authentication	Password	AP Group	LWAP
Admin_A	50	WPA2 Enterprise / 802.1X	RADIUS User Account	Group A	LWAP_A
Employees_A	150	WPA2 / PSK	employees123		
Admin_B	60	WPA2 Enterprise / 802.1X	RADIUS User Account	Group B	LWAP_B
Employees_B	160	WPA2 / PSK	employees123		

Wireless Connectivity Verification with RADIUS

